

Spheroidal Harmonic Analysis

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1 Introduction

Spheroidal objects arise in many physical contexts ranging from astrophysics [1] to microbiology [2] and electrostatics [3]. Spheroidal geometries provide a generalization of spherical geometries that maintains several useful properties spherical geometries provide.

All of the applications mentioned above require solving ellipsoidal partial differential equations on the spheroidal surface. These equations may be scalar, as in the case of gravitation, or vector, as arises in Stokes problems. Both types of problems are commonly solved by employing layer potentials, which may be used to reformulate the differential equations as second kind integral equations on the particle boundary. Layer potential methods are not without their own set of difficulties. They require the evaluation of integrals with singular kernels on particle surfaces. Since the integral equations incorporate pairwise interactions between all particles, the number of integrals that must be computed grows quickly as the number of surfaces increases. Previous works have worked to solve these problems in the case of spherical particles by spherical harmonic methods, demonstrating that the spherical harmonics diagonalize the single and double layer operators on a sphere with resulting spectral values that can be computed analytically [4]. This analysis forms the crux of a method for efficient evaluation of scalar and vector potentials on and near the surface of spheres while avoiding the need for the integration of singular integrals. This method may be combined with the Fast Multipole Method for interactions between distance bodies to develop a highly scalable evaluation routine [5]. This routine forms the basis of a method for solving boundary value problems on particle surfaces that scales linearly with the number of particles.

Spheroidal harmonics form an orthogonal basis for functions on spheroidal surfaces and are occasionally used as basis functions for Galerkin methods involving spheroids [6]. Our contribution is to follow the methods used for the spherical case to develop a near evaluation routine on spheroids. We demonstrate that the double layer operator diagonalizes in the spheroidal harmonic basis and that a modified basis can be used to similar effect for the single layer. We use these properties to develop a near evaluation routine for evaluations of layer operators and their derivatives on and near the surface of spheroids which converges spectrally without need for evaluating singular integrals.

Finally, we show how these may be used to compute a Stokes problem on the surface of a spheroid. Analytical solutions for the Stokes equation for ellipsoidal particles [7] and in spheroidal coordinates [8] can be derived. However, such calculations cannot be used for multiple particle systems. We demonstrate how layer potentials may be used to compute numerical solutions to Stokes systems involving spheroidal particles.

2 Prolate Spheroidal Coordinates

We begin with an overview of prolate spheroidal coordinates. Prolate spheroidal coordinates are a family of orthogonal coordinates which vary with a parameter, a . The prolate spheroids of a particular spheroidal coordinate system have semifocal length a . Although we limit the following discussion to prolate spheroidal coordinates, analogous formulas exist for oblate spheroidal coordinate systems. Greater care must be taken with those systems however, as there is not a one-to-one correspondence between them and Cartesian coordinates.

2.1 Coordinates

There are many different ways of writing down coordinate maps in spheroidal coordinates ¹. We use the following:

$$x = a\sqrt{u^2 - 1}\sqrt{1 - v^2} \cos \phi \quad (2.1)$$

$$y = a\sqrt{u^2 - 1}\sqrt{1 - v^2} \sin \phi \quad (2.2)$$

$$z = auv \quad (2.3)$$

Prolate spheroidal coordinates are of the form (u, v, ϕ) , with $u > 1$, $-1 \leq v \leq 1$, and $0 \leq \phi \leq 2\pi$. Spheroidal parametric surfaces are obtained by holding u constant, while holding v constant parametrizes hyperboloids of 2 sheets. Cross sections of spheroids are plotted in figure 1, which illustrates how the spheroids become closer to spheres as u increases.

The inverse coordinate map is given by

$$u = \frac{1}{2a} \left(\sqrt{x^2 + y^2 + (z + a)^2} + \sqrt{x^2 + y^2 + (z - a)^2} \right) \quad (2.4)$$

$$v = \frac{1}{2a} \left(\sqrt{x^2 + y^2 + (z + a)^2} - \sqrt{x^2 + y^2 + (z - a)^2} \right) \quad (2.5)$$

$$\phi = \arctan y/x \quad (2.6)$$

We note that the quantities $\sqrt{x^2 + y^2 + (z \pm a)^2}$ are the distance between point (x, y, z) and the two focal points of the spheroid. Hence au is average distance to the focal points and av is half the difference between the distances. As is the case in spherical coordinates, $\arctan$ is defined so that its range is $(-\frac{\pi}{2}, \frac{3\pi}{2})$, for instance Matlab's `atan2` function.

The surface area of a spheroid is given by:

$$SA = 2\pi b^2 \left(1 + \frac{c \arcsin(\epsilon)}{a\epsilon} \right) \quad (2.7)$$

where $c = au$ is the semi-major axis, $b = a\sqrt{u^2 - 1}$ is the semi-minor axis and $\epsilon = \sqrt{1 - \frac{b^2}{c^2}} = \frac{1}{u}$ is the eccentricity of the generating ellipse.

Prolate spheroidal coordinates are sometimes written with coordinates (μ, τ, ϕ) where $u = \cosh \mu$ and $v = \cos \tau$. Substituting these coordinates in to equation 2.1 shows more clearly the connection between spherical and spheroidal coordinates. We choose the (u, v, ϕ) formulation for this paper for notational convenience.

2.2 Differential Operators in Spheroidal Coordinates

Spheroidal coordinates allow us to parametrize a spheroid of given u as

$$\vec{x}(v, \phi) = (a\sqrt{u^2 - 1}\sqrt{1 - v^2} \cos \phi, a\sqrt{u^2 - 1}\sqrt{1 - v^2} \sin \phi, auv)$$

From this we obtain expressions for the tangent and normal vectors:

$$\begin{aligned} \vec{x}_v &= \left(-a\sqrt{\frac{u^2 - 1}{1 - v^2}}v \cos \phi, -a\sqrt{\frac{u^2 - 1}{1 - v^2}}v \sin \phi, au \right) \\ \vec{x}_\phi &= \left(-a\sqrt{u^2 - 1}\sqrt{1 - v^2} \sin \phi, a\sqrt{u^2 - 1}\sqrt{1 - v^2} \cos \phi, 0 \right) \\ \vec{u} = \vec{x}_v \times \vec{x}_\phi &= \left(a^2u\sqrt{u^2 - 1}\sqrt{1 - v^2} \cos \phi, a^2u\sqrt{u^2 - 1}\sqrt{1 - v^2} \sin \phi, a^2(u^2 - 1)v \right) \end{aligned} \quad (2.8)$$

¹See also: https://en.wikipedia.org/wiki/Prolate_spheroidal_coordinates

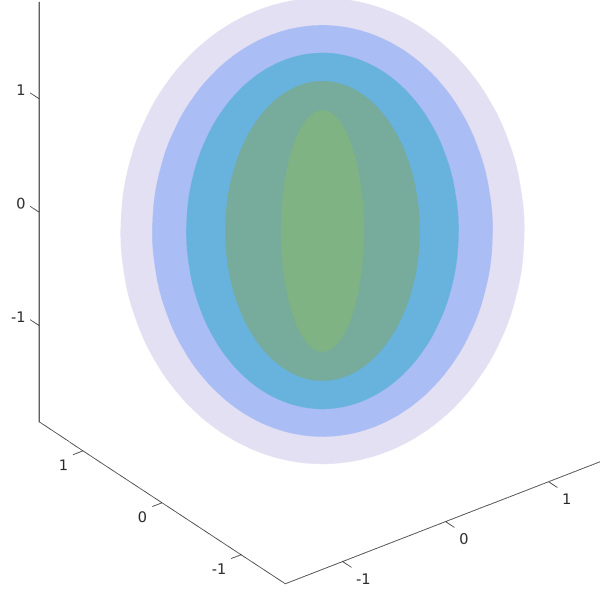


Figure 1: Spheroids of constant u in the spheroidal coordinate system with $a = 1$. u ranges from 1.05 to 2. As u approaches 1, the ellipse becomes more eccentric.

computing the magnitude of the quantities in equation 2.8 yields scale factors:

$$h_u = a\sqrt{\frac{u^2 - v^2}{u^2 - 1}}, \quad h_v = a\sqrt{\frac{u^2 - v^2}{1 - v^2}}, \quad h_\phi = a\sqrt{u^2 - 1}\sqrt{1 - v^2} \quad (2.9)$$

orthonormal frames on spheroidal surfaces are given via these quantities:

$$e_v = \frac{\vec{x}_v}{h_v}, \quad e_\phi = \frac{\vec{x}_\phi}{h_\phi}, \quad e_u = \frac{\vec{x}_v \times \vec{x}_\phi}{\|\vec{x}_v \times \vec{x}_\phi\|} \quad (2.10a)$$

The gradient operator is then given by:

$$\nabla f = \frac{1}{h_v} \frac{\partial f}{\partial v} e_v + \frac{1}{h_\phi} \frac{\partial f}{\partial \phi} e_\phi + \frac{1}{h_u} \frac{\partial f}{\partial u} e_u \quad (2.11)$$

2.3 Spheroidal Laplacian

The Laplacian operator in prolate spheroidal coordinates is given by

$$\nabla^2 = \frac{1}{a^2(u^2 - v^2)} \left(\frac{\partial}{\partial u} \left[(u^2 - 1) \frac{\partial}{\partial u} \right] + \frac{\partial}{\partial v} \left[(1 - v^2) \frac{\partial}{\partial v} \right] \right) + \frac{1}{a^2(u^2 - 1)(1 - v^2)} \frac{\partial^2}{\partial \phi^2} \quad (2.12)$$

Like the spherical Laplacian, the spheroidal Laplacian is separable, with solution of the forms

$$P_n^m(u) P_n^m(v) e^{im\phi} \quad (2.13a)$$

in the interior of the spheroid and

$$Q_n^m(u) P_n^m(v) e^{im\phi} \quad (2.13b)$$

in the exterior.

Where P_n^m, Q_n^m are the Legendre polynomials of the first and second kind. These solutions are known as *Spheroidal Harmonics*. Figure 2 plots the real part of a few lower order harmonics. $P_n^m(u)$ and $Q_n^m(u)$ differ from the Legendre functions encountered in spherical harmonics in that they must

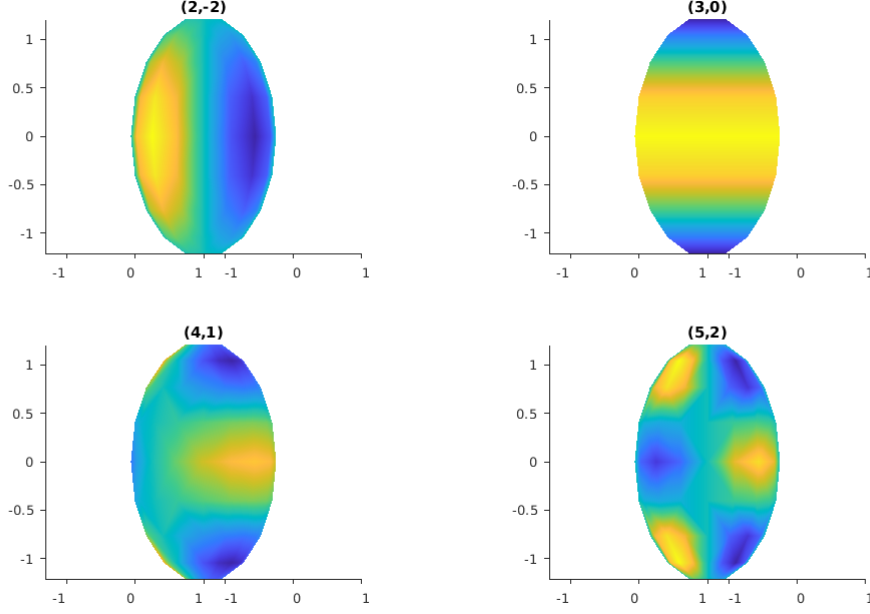


Figure 2: Examples of spheroidal harmonics

be evaluated for $u > 1$. Their domains are extended by defining them to be solutions to the Legendre equation:

$$|1 - x^2| \frac{d^2}{dx^2} f(x) - 2x \frac{d}{dx} f(x) + \left[n(n+1) - \frac{m^2}{|1 - x^2|} \right] f(x) = 0 \quad (2.14)$$

This extension preserves the recurrence relationships between the functions. We employ a backwards stable algorithm described in [9].

Surface Harmonics On the surface of a spheroid with fixed u , we obtain *Spheroidal Surface Harmonics* of the form

$$S_n^m = P_n^m(v) e^{im\phi} \quad (2.15)$$

The S_n^m form a basis for L^2 functions on the surface of a spheroid and are orthogonal with respect to weight function

$$w_u(v) = \frac{1}{a^2(u^2 - v^2)(u^2 - 1)}$$

[10]. We then have that

$$\int S_{n_1}^{m_1} S_{n_2}^{m_2} dS = \int_0^{2\pi} \int_{-1}^1 S_{n_1}^{m_1} S_{n_2}^{m_2} dv d\phi = \gamma_n^m \delta_{n_1, n_2} \delta_{m_1, m_2} \quad (2.16)$$

with normalization constant γ_n^m given by

$$\gamma_n^m = \frac{2(n+m)!}{(2n+1)(n-m)!} (1 + \delta_{0,m}) \pi \quad (2.17)$$

Unless otherwise stated, we normalize S_n^m with $\sqrt{\gamma_n^m}$ so that the inner product of S_n^m with itself is 1.

Just as spherical harmonics form an orthonormal basis for $L^2(S^2)$, spheroidal harmonics form an

orthonormal basis for functions on the surface of a spheroid S_u . We have that

$$f(v, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm} S_n^m \quad (2.18)$$

with $a_n^m = \int_{S_u} f S_n^m dS_u = \int_0^{2\pi} \int_{-1}^1 f(v, \phi) S_n^m(v, \phi) w_u(v) dv d\phi$

2.4 General Ellipsoidal Coordinates

We end this section by briefly highlighting the fact that spheroidal coordinates are a particular type of *Ellipsoidal Coordinates*, a system of orthonormal coordinates that have ellipsoids with no assumed symmetry as level surfaces [11]. The ellipsoidal Laplace equation yields *ellipsoidal harmonics*, which are employed in many of the same applications that spheroidal harmonics are, such as electrostatics [12]. Ellipsoidal harmonics are constructed out of *Lamé functions*. Lamé functions lack the recurrence relations found in Legendre functions and can be expressed in terms of elementary functions only for small m . For these reasons they are both far less amenable to analysis and far more expensive to evaluate numerically.

3 Layer Operators

Differential equations can be recast as boundary integral equations by employing layer potentials, which are used to form well conditioned second kind integral equations. It can be shown that spherical harmonics are eigenfunctions for the single and double layer operators on a sphere with corresponding eigenvalues that can be computed analytically [13]. This property forms the basis of a fast layer potential evaluation technique [4] that can be used for both scalar and vector potentials in problems involving spheres. In this section we extend that result to prolate spheroids. We shall show that prolate spheroidal harmonics are eigenfunctions of the double layer operator.

3.1 Definitions

Layer operators are special integral operators that are useful for expressing solutions to ellipsoidal PDEs in terms of well conditioned integral equations. The single and double layer potentials have kernels in terms of Green's functions. For the Laplace equation, the Greens function is given by $G(x, y) = \frac{1}{\|x - y\|}$. We have the single layer operator given by

$$S[\sigma](x, y) = \int_{\Gamma} \frac{1}{\|x - y\|} \sigma(y) dS(y) \quad (3.1)$$

and the double Layer operator defined as

$$D[\sigma](x, y) = \int_{\Gamma} \frac{\partial}{\partial \nu} \frac{1}{\|x - y\|} \sigma(y) dS(y) \quad (3.2)$$

The operator kernels may be expressed in terms of spheroidal coordinates with respect to a point y on the surface of a spheroid. Let $x = (x_1, x_2, x_3) = (u_x, v_x, \phi_x)$, $y = (y_1, y_2, y_3) = (u_y, v_y, \phi_y)$ with $y \in \Gamma$. We may express each term of $r = x - y$ in spheroidal coordinates. For example,

$$r_1(u_x, v_x, \phi_x, u_y, v_y, \phi_y) = x_1 - y_1 = \sqrt{u_x^2 - 1} \sqrt{1 - v_x^2} \cos \phi_x - \sqrt{u_y^2 - 1} \sqrt{1 - v_y^2} \cos \phi_y$$

And likewise for $(x_2 - y_2), (x_3 - y_3)$, which we denote by r_2, r_3 respectively. From this, the single layer kernel is readily expressed. The double layer operator may then be written

$$\frac{\partial}{\partial \nu} \left(\frac{1}{\|x - y\|} \right) = \frac{r_1 \frac{\partial r_1}{\partial u_y} + r_2 \frac{\partial r_2}{\partial u_y} + r_3 \frac{\partial r_3}{\partial u_y}}{\|r\|^{3/2}} \quad (3.3)$$

3.2 Double Layer Operator in Spheroidal Coordinates

We will see that the double layer operator yields a more simple expression in terms of spheroidal harmonics, so we describe it first. We show that the spheroidal harmonics are in fact eigenvectors of the double layer operator. Following the analysis on the sphere done in [13], using the jump conditions of the double layer operator to obtain expressions for the interior and exterior spherical harmonic coefficients.

Let $\sigma(v, \phi)$ be a function on the surface of a spheroid. To spheroidal harmonic order p , σ is given by

$$\sigma(v, \phi) = \sum_{n,m} \sigma_{nm} S_n^m(v, \phi) \quad (3.4)$$

We write the double layer operator operator of σ as

$$D[\sigma](v, \phi) = \sum_{n,m} a_{nm}^i P_n^m(u) S_n^m(v, \phi) \quad (3.5)$$

in the interior and

$$D[\sigma](v, \phi) = \sum_{n,m} a_{nm}^e Q_n^m(u) S_n^m(v, \phi) \quad (3.6)$$

The double layer operator has the jump conditions $[[D\sigma]] = -\sigma$, $[[\frac{\partial}{\partial \nu} D\sigma]] = 0$. Subtracting the exterior expressions of $D[\sigma](u, v)$ from the interior expression, making use of the jump condition and orthogonality of the S_n^m on the surface of the spheroid, we find that

$$a_{nm}^i P_n^m(u) - a_{nm}^e Q_n^m(u) = -\sigma_{nm} \quad (3.7)$$

To solve for a_{nm}^i, a_{nm}^e we find a second equation relating them by using the jump condition in the normal derivative. The normal derivative operator in (u, v, ϕ) coordinates is given by

$$\frac{1}{\sqrt{\frac{u^2-v^2}{u^2-1}}} \frac{\partial}{\partial u}$$

but, since we have that that

$$\frac{1}{\sqrt{\frac{u^2-v^2}{u^2-1}}} \frac{\partial}{\partial u} \left[\sum_{nm} a_{nm}^i P_n^m(u) S_n^m(v, \phi) - \sum_{nm} a_{nm}^e Q_n^m(u) S_n^m(v, \phi) \right] = 0$$

and $u^2 - v^2 > 0$ we can obtain the relationship

$$a_{nm} P_n^{m'}(u) - a_{nm}^e Q_n^{m'}(u) = 0 \quad (3.8)$$

We thus have a 2×2 linear equation to solve for a_{nm}^e and a_{nm}^i in terms of σ_{nm} . Solving this equation yields

$$a_{nm}^i = \frac{\sigma_{nm} Q_n^{m'}(u)}{P_n^m(u) Q_n^{m'}(u) - P_n^{m'}(u) Q_n^m(u)} \quad (3.9)$$

$$a_{nm}^e = \frac{\sigma_{nm} P_n^{m'}(u)}{P_n^m(u) Q_n^{m'}(u) - P_n^{m'}(u) Q_n^m(u)} \quad (3.10)$$

Finally, we may employ the Wronskian relation

$$W(P_n^m, Q_n^m) = \frac{(n+m)! (-1)^{m+1}}{(n-m)! u^2 - 1}$$

to obtain

$$a_{nm}^i = \frac{(n-m)!}{(n+m)!} (-1)^m (u^2 - 1) \sigma_{nm} Q_n^{m'}(u) \quad (3.11)$$

$$a_{nm}^e = \frac{(n-m)!}{(n+m)!} (-1)^m (u^2 - 1) \sigma_{nm} P_n^{m'}(u) \quad (3.12)$$

This proves the following

Theorem 1. *Let S_n^m be the surface spheroidal harmonics of a spheroid of parameter u . Denote the closure of the spheroid as Ω and its surface as Γ . Let a_{nm}^i, a_{nm}^e be as above with $\sigma_{nm} = 1$. Then we have the following:*

1. $\mathcal{D}[S_n^m](x)|_\gamma = \lambda_{nm} S_n^m$ with $\lambda_{nm} = \frac{1}{2}(a_{nm}^i + a_{nm}^e)$
where a_{nm}^e, a_{nm}^i are given above, with $\sigma_{mn} = 1$.
2. $\mathcal{D}[S_n^m](x) = \begin{cases} a_{nm}^e Q_n^m(u_x) & x \in \Omega^c \\ a_{nm}^i P_n^m(u_x) & x \in \Omega \end{cases}$

Completion Field for Double Layer The discussion above shows that $\mathcal{D}$ is signalized by the spheroidal harmonics. We will show in the next section that $\mathcal{S}$ is not so easily diagonalized. Therefore, it may be desirable to formulate Laplace systmes in terms of $\mathcal{D}$ alone. It is well known that $\mathcal{D}$ has a one dimensional kernel for exterior problems. We eliminate this kernel by adding the standard double layer completion field, $\mathcal{N}$:

$$\mathcal{N}[\sigma](x) = Q_{00}(u_x) Y_{00} \sigma_{00} \quad (3.13)$$

Given a surface label, f , we may then solve for a density σ by inverting

$$(D + \mathcal{N} + 0.5I) [\sigma] = f$$

any evaluating

$$(D + \mathcal{N})[\sigma]$$

at points in the exterior.

We validate the formulas above and demonstrate their use by using them to solve a simple Laplace problem. We place a single point charge in the interior of a spheroid and use it to compute the potential on the spheroidal surface, f . Then we solve the boundary value problem

$$\nabla^2 \phi(x) = 0 \quad x \notin \Gamma \quad \phi = f \quad x \in \Gamma \quad (3.14)$$

by constructing the boundary integral equation

$$\phi(x) = (D + \mathcal{N})[\mu](x) \quad x \notin \Gamma \quad (3.15)$$

$$\phi(x) = \left(\frac{1}{2}I + D + \mathcal{N}\right)[\mu](x) \quad x \in \Gamma \quad (3.16)$$

equation 3.16 is a second kind integral equation, which we solve using GMRES, applying $\mathcal{D}$ and $\mathcal{N}$ via their spectral representation.

3.3 Single Layer Operator

Although we approach the single layer operator in an identical fashion to the double layer, the analysis fails to be as simple. This is because the normal derivative contains a non-separable factor of $\frac{1}{\sqrt{u^2 - v^2}}$,

which cannot be removed because the normal derivative jump condition for $\mathcal{S}$ is nonzero. To solve for a_{nm}^i and a_{nm}^e in this case, we have

$$\begin{bmatrix} P_n^m(u) & Q_n^m(u) \\ P_n^{m'}(u) & Q_n^{m'}(u) \end{bmatrix} \begin{bmatrix} a_{nm}^i \\ a_{nm}^e \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{1-u^2}} \tilde{\sigma}_{nm} \end{bmatrix} \quad (3.17)$$

where $\tilde{\sigma}_{nm}$ is the Spheroidal Harmonic transform of $\sqrt{u^2 - v^2} \sigma(v, \phi)$. The spheroidal harmonic coordinates are then:

$$a_{nm}^i = \frac{(n-m)!}{(n+m)!} (-1)^{m+1} \sqrt{u^2 - 1} \tilde{\sigma}_{nm} Q_n^m(u) \quad (3.18a)$$

$$a_{nm}^e = \frac{(n-m)!}{(n+m)!} (-1)^{m+1} \sqrt{u^2 - 1} \tilde{\sigma}_{nm} P_n^m(u) \quad (3.18b)$$

This computation has been verified numerically and can be interpreted in the following way:

Theorem 2 (Single Layer). *The Single Layer Operator on a Spheroid, $\mathcal{S}$, maps $\frac{S_n^m(v, \phi)}{\sqrt{u^2 - v^2}}$ to $\alpha_{nm} Y_n^m(v, \phi)$ with α_{nm} given by the expressions in equation 3.18a in the interior and 3.18b in the exterior.*

Change of Basis Let $\mathcal{G}_u : L_2(\Gamma) \rightarrow L_2(\Gamma)$ be the operator defined such that

$$\mathcal{G}_u[f](v, \phi) = \frac{f(v, \phi)}{\sqrt{u^2 - v^2}} \quad (3.19)$$

Since $u > 1$ and $\|v\| \leq 1$, this operator is continuous.

It is straightforward to show that the $\frac{Y_n^m}{\sqrt{u^2 - v^2}}$ form another basis for the same space as Y_n^m . We define the matrix G_u to be the change of basis matrix between the orthogonal basis $\mathcal{B} = \{Y_{nm}\}$ and the basis $\mathcal{B}_u = \{\frac{Y_{nm}}{\sqrt{u^2 - v^2}}\}$. In other words, we have that

$$(G_u)_{ij} = \langle Y_i, \frac{Y_j}{\sqrt{u^2 - v^2}} \rangle$$

, where i and j correspond to (n_i, m_i) and (n_j, m_j) in the standard ordering.

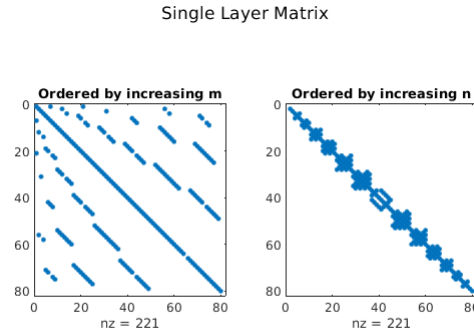


Figure 3: Single Layer matrix nonzero entries

In figure 3 we plot the nonzero entries of the $\mathcal{S}$ matrix. Unlike the $\mathcal{D}$ matrix, this matrix is not diagonal. However, plotting the entries in increasing m order reveals that the non-zero entries are clustered along the diagonal, forming “bulges” of size n . We propose the following subroutine to evaluate $\mathcal{S}[\sigma]$ on a spheroid of given u :

1. compute the spheroidal coordinates, $\hat{\sigma}$, via the spheroidal harmonic transform.
2. compute the coordinates in basis $\mathcal{B}_u$: $\hat{\sigma}_u = G_u^{-1}\hat{\sigma}$
3. apply $\mathcal{S}$ via the formulas given in equation 3.18b.
4. apply the Spherical Harmonic Syn function to recover $\mathcal{S}[\sigma]$ in spacial coordinates.

We solve the same point charge problem to verify the accuracy of this subroutine. In this case, we solve

$$\mathcal{S} + \mathcal{D} + \frac{1}{2}I$$

at the boundary. The results are shown in figure 4. We observe similar overall accuracy as for the $\mathcal{D} + \mathcal{N}$ system. The principle difference between the two systems is the need to invert G_u , a sparse and well conditioned matrix.

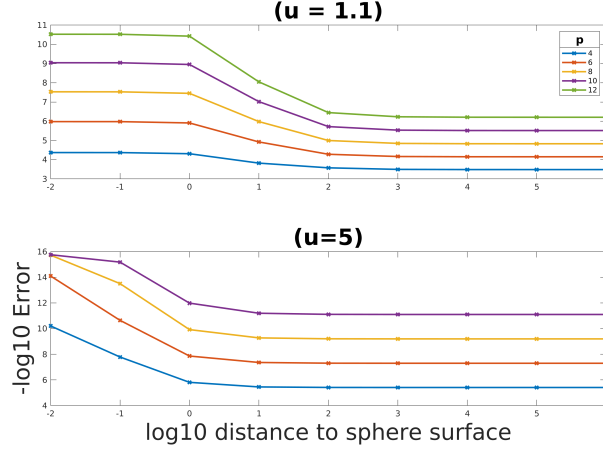


Figure 4: Error for S+D point charge system. When u is close to 1 (top) convergence is slower. For larger u (bottom) convergence is more rapid.

3.4 Derivative Computation

To solve Laplace problems, we will need to express the gradient and normal derivative of potentials. Defining e_v, e_ϕ, e_u as in equations 2.10 we have that

$$\nabla f = \frac{1}{a\sqrt{\frac{u^2-v^2}{1-v^2}}} \frac{\partial f}{\partial v} e_v + \frac{1}{a\sqrt{u^2-1}\sqrt{1-v^2}} \frac{\partial f}{\partial \phi} e_\phi + \frac{1}{a\sqrt{\frac{u^2-v^2}{u^2-1}}} \frac{\partial f}{\partial u} e_u \quad (3.20)$$

Further we will denote by ∇_Γ the first two terms of the above sum, which are to projection of the gradient onto the tangent space of the spheroid.

To express ∇f in terms of spheroidal harmonics, we first find expressions for the partial derivatives in spheroidal coordinates. Letting $f = \sum a_n^m Y_n^m(v, \phi) Q_n^m(u) = \sum a_n^m e^{im\phi} P_n^m(v) Q_n^m(u)$, we have that

$$f_v = \sum a_n^m Q_n^m(u) \frac{1}{(1-v^2)} [-(n+1)vY_n^m + (n-m+1)Y_{n+1}^m] \quad (3.21)$$

$$f_\phi = \sum (ima_n^m) Y_n^m(v, \phi) Q_n^m(u) \quad (3.22)$$

$$f_u = \sum a_n^m Y_n^m(v, \phi) Q_n^{m'}(u) \quad (3.23)$$

Thus we get that

$$\nabla_{\Gamma} f = \sum a_{nm} Q_n^m \left(\frac{1}{(1-v^2)} [-(n+1)vY_n^m + (n-m+1)Y_{n+1}^m] e_v + \frac{im}{\sqrt{(u^2-1)(1-v^2)}} Y_n^m e_{\phi} \right) \quad (3.24)$$

∇_{Γ} is not diagonalized in the harmonic basis but is rather tridiagonal. Likewise, we have that:

$$e_u^T \nabla f = \sqrt{\frac{u^2-1}{u^2-v^2}} \sum a_{nm} Y_n^m(v, \phi) Q_n^{m'}(u) \quad (3.25)$$

Both of these values can be computed from the spheroidal values, either directly or via a modification similar to that discussed in the computation of $\mathcal{S}$.

3.5 S' Operator

We may combine the conclusion of theorem 2 with the representation above for the normal derivative to obtain the following:

Theorem 3. *The spheroidal harmonics diagonalize the $\mathcal{S}'$ operator*

Proof. Consider the exterior harmonics at a point with $u > u_0$ the surface parameter of the spheroid. We have $\mathcal{S}[S_n^m] = a_n^m \sqrt{u_0^2 - v^2} S_n^m Q_n^m(u)$. By 3.20, the normal derivative is then $\mathcal{S}'[S_n^m] = a_n^m \sqrt{u_0^2 - 1} Q_n^{m'}(u) S_n^m$. The interior harmonic follows. $\square$

We summarize the results as follows: Let u_0 be the constant surface parameter of spheroid Γ

Potential	Interior	Exterior
$\mathcal{D}[S_n^m]$	$c(-1)^m(u_0^2 - 1)Q_n^{m'}(u_0)P_n^m(u)$	$c(-1)^m(u_0^2 - 1)P_n^{m'}(u_0)Q_n^m(u)$
$\mathcal{S}[\frac{S_n^m}{\sqrt{u^2-v^2}}]$	$-c\sqrt{u_0^2 - 1}Q_n^m(u_0)P_n^m(u)$	$-c\sqrt{u_0^2 - 1}P_n^m(u_0)Q_n^m(u)$
$\mathcal{S}'[S_n^m]$	$c(-1)^m(u_0^2 - 1)Q_n^m(u_0)P_n^{m'}(u)$	$c(-1)^m(u_0^2 - 1)P_n^m(u_0)Q_n^{m'}(u)$

where $c = \frac{(n-m)!}{(n+m)!}$.

3.6 Stokes

We may combine the $\mathcal{S}$ operator and its derivatives to obtain a method for evaluating the Stokeslet solution. We follow the formula in [14], which has the Stokeslet of surface vector density $\mathbf{f} = (f_1, f_2, f_3)$ as:

$$(S[\mathbf{f}])_i = \sum_{j=1}^3 \left[\left(\delta_{ij} - \mathbf{x}_j \frac{\partial}{\partial x_i} \right) \mathcal{S}[f_j] \right] + \frac{\partial}{\partial x_i} \mathcal{S}[\mathbf{x} \cdot \mathbf{f}] \quad (3.26)$$

We may use this to evaluate Stokes single layer operators ($\mathcal{S}[\mathbf{f}]$) on the surface of spheroids, solving for the fluid in the exterior. We present cross sections of the flows that occur for constant background flows in the e_1 and e_3 directions. The resulting system was solved using 20 gmres iterations to achieve a residual error of $1e-8$.

4 Explicit Mapping to and from Spherical

For the sake of computational convenience, we note here explicit conversions between prolate spheroidal coordinates and spherical coordinates (ρ, θ, ϕ) . Using the Cartesian definition, it is straightforward to show that

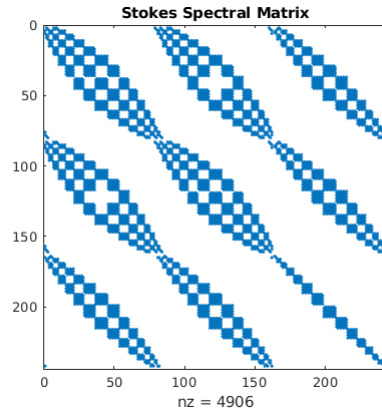


Figure 5: Nonzero entries of Spectral mapping, $p=8$

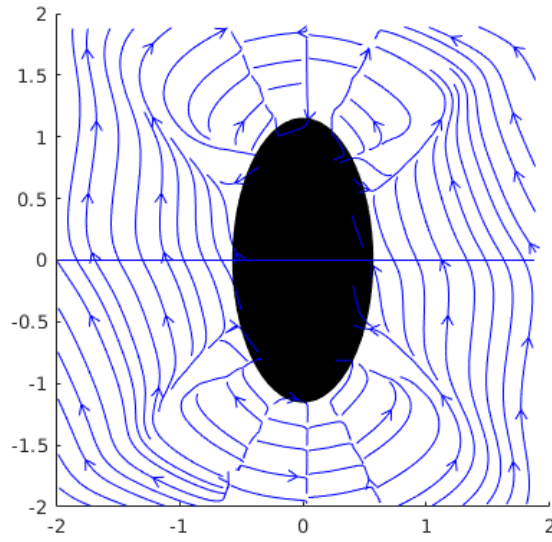


Figure 6: Fluid field around spheroid of $u = 1.15$. The background flow is constant $[0;0;1]$ with a no slip condition on the spheroidal surface

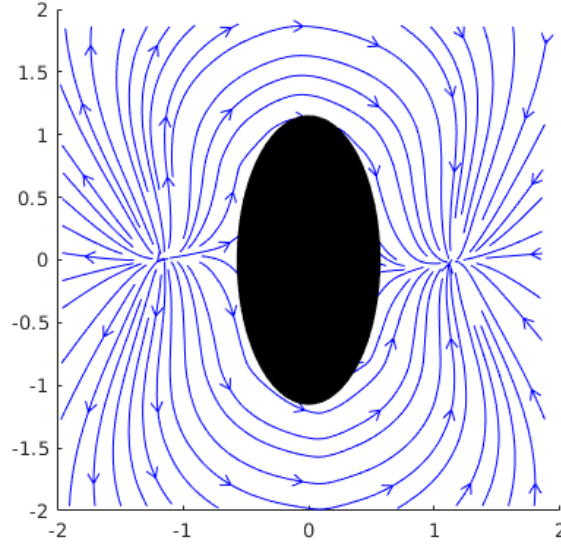


Figure 7: Fluid field around spheroid of $u = 1.15$. The background flow is constant $[1;0;0]$ with a no slip condition on the spheroidal surface

$$\rho = \sqrt{x^2 + y^2 + z^2} = a\sqrt{u^2 + v^2 - 1} \quad (4.1)$$

$$\theta = \arctan \frac{\sqrt{x^2 + y^2}}{z} = \arctan \frac{\sqrt{u^2 - 1}\sqrt{1 - v^2}}{uv} \quad (4.2)$$

Working backwards, we observe from the expression of ρ in spheroidal coordinates that

$$u = \frac{1}{2} \left[\left(\sqrt{\frac{\rho^2 + 2a\rho \cos \theta}{a^2} + 1} \right) + \left(\sqrt{\frac{\rho^2 - 2a\rho \cos \theta}{a^2} + 1} \right) \right] \quad (4.3)$$

$$v = \frac{1}{2} \left[\left(\sqrt{\frac{\rho^2 + 2a\rho \cos \theta}{a^2} + 1} \right) - \left(\sqrt{\frac{\rho^2 - 2a\rho \cos \theta}{a^2} + 1} \right) \right] \quad (4.4)$$

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